

DATA612 Final Project Proposal

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Movie Recommendation System Using MovieLens 25M Dataset

Dataset

For my final project, I will use the MovieLens 25M dataset provided by GroupLens. The dataset will contain approximately 25 million ratings from over 162,000 users on more than 62,000 movies. It will also include additional information such as movie titles, genres, and user-generated tags. The large size of the dataset will make it appropriate for building and evaluating a scalable recommender system.

Data Preparation

The dataset will be explored and cleaned before building the recommender system. Missing or unnecessary data will be removed if needed, and the ratings data will be transformed into the format required by Spark ALS. I will also examine the distribution of users, movies, and ratings to better understand the dataset before training the model.

Exploratory Analysis

Several visualizations will be created to better understand the data, including the distribution of movie ratings, the number of ratings per user, and the most frequently rated movies and genres. This analysis will help identify patterns within the dataset before model development.

Model

I plan to build a collaborative filtering recommender system using the Alternating Least Squares (ALS) algorithm in PySpark. Different ALS parameters, such as rank, regularization, and the number of iterations, will be tested to determine which combination produces the best recommendations.

Evaluation

The dataset will be divided into training and testing sets using an 80/20 split. Model performance will be evaluated primarily using Root Mean Squared Error (RMSE). After

selecting the best-performing model, I will generate Top-N movie recommendations for selected users.

Unique Element

In addition to generating recommendations, I will incorporate movie genres and tags to provide more meaningful explanations for why particular movies are recommended. This will improve the interpretability of the recommender system beyond simply predicting ratings.

Expected Outcome

The completed project will demonstrate how Spark ALS can efficiently build a collaborative filtering recommender system using a large-scale dataset. The final system should provide personalized movie recommendations while illustrating the concepts of recommendation modeling, parameter tuning, and model evaluation learned throughout this course.